Binomial Distribution

a discrete r.v is said to follow Binomial distribution if the pmf of X is as follows

$p(X=k)={}^{n}C_{k}\,p^{k}q^{n-k}$, k=0,1,2,...,n and p+q=1. It is denoted as $X \sim B(n,p,k)$,

where n and p are two parameters.

The Binomial distribution holds under the following conditions:

1. The number of trial, i.e., n is constant.

2. Trails are independent, i.e., result of one trail will not effect the result other trail

3. Only two things can happen, i.e., either a success will occur or a failure will occur. This implies that the outcomes will be only two either 0 or 1, either head will occur or tail will occur, etc.

4. Probability of success is constant in each and every trail and it is equal to p.

Properties of Binomial distribution:

If X_1 and X_2 are two independent Binomial variate with parameters n_1, p and n_2, p respectively then X_{1+} X_2 a Binomial distribution with parameter $(n_{1+} n_2)$, p. It can be extend for n number of independent Binomial variate. This is known as additive property of Binomial distribution.

The mean of binomial distribution is ***np*** and the variance is ***npq,*** and ***npq< np***

The variance of Binomial distribution attains maximum at p=q=0.5 and the maximum value is n/4.

mean $\mu' = E(X) = \sum_{k} k \times prob(X=k) = \sum_{k} k\,{}^{n}C_{k}\,p^{k}q^{n-k} = np$, Similarly variance is

$\sigma^2 = E(X^2) - \{E(X)\}^2 = npq$

The moment generating function is $M_x(t) = \sum_{k} e^{tk}\,{}^{n}C_{k}\,p^{k}q^{n-k} = (pe^{t}+q)^{n}$

Therefore mean

$E(X) = M_x'(t)\; at\; t=0$ is np simillary variance is $\sigma^2 = E(X^2) - \{E)X)\}^2 = M_x''(t) - \{M_x'(t)\}^2\; t=0\; npq$

Examples

1. mean of a binomial distribution is 3 and variance is 4 is it true?

2. Mean and variance of a Binomial distribution is 4 and 4/3 respectively. find prob(x>=1)

3. A contractor purchase a shipment of 100 transistors. If it is his policy to test these transistors and keep the shipment iff at test 9 out of 10 are in working conditions. if the shipment contains 20 defective transistors, what is the probability that it will be kept.

4. In a precision bombing attack there is a 50% chance that a bomb will hit the target. If two direct hits are required to destroy the target completely. how many bomb must be dropped to give a 99% chance or better to completely destroy the target?